

Subsurface damage evolution and material removal mechanism of monocrystalline silicon by in-plane ultrasonic vibration-assisted scratching: An experimental study

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Abstract

The enviable rise in demand for high-precision monocrystalline silicon places emphasis on achieving near-damage-free manufacturing. Conventional methods are under scrutiny due to an increasing concern over their susceptibility to brittle fracture, which easily induces severe subsurface damage (SSD). Modern engineering requires high surface integrity, where such defects must be minimized. Ultrasonic vibration-assisted machining (VAM) has been proven to facilitate ductile removal in various brittle materials. However, its specific mechanisms for monocrystalline silicon remain ill understood. Because silicon undergoes complex phase transformations under varying stress states, how high-frequency kinematics alter these transitions is underrepresented in current literature. Existing models often miss these transient stress variations, thereby limiting performance optimization for large-scale engineering.

This study investigates SSD evolution and material removal in silicon via in-plane VAM scratching (~103 kHz). By comparing non-ultrasonic and varying UV amplitudes (10 nm, 50 nm, 110 nm), this work provides a detailed reference for dynamic tool-material interactions. As evidenced by SEM and TEM, optimal parameters (e.g., 50 nm) suppress brittle fracture, allowing the material to better withstand processing stresses with significantly reduced damage. Furthermore, the evolution of the amorphous layer and phase transformations is discussed in detail, revealing how intermittent cutting facilitates dynamic stress relief. The impact of these findings establishes a framework for minimizing SSD and improving machining precision, which will find use in diverse and wide applications across advanced microelectronics and optics industries.

Keywords: Monocrystalline silicon, Ultrasonic vibration-assisted machining, Subsurface damage, Material removal mechanism

1. Introduction

The semiconductor and advanced optoelectronics industries are currently experiencing an unprecedented rise in demand for high-precision monocrystalline silicon components[1–5]. As the foundational material for next-generation integrated circuits, infrared optical lenses, and micro-electromechanical systems (MEMS), the functional reliability of silicon is intrinsically linked to its surface and subsurface quality[6,7]. Consequently, modern ultra-precision manufacturing places emphasis on achieving near-damage-free surfaces, as even nanoscale defects can drastically degrade the electrical and optical performance of the final devices[8].

However, the inherent high hardness and severe brittleness of monocrystalline silicon present formidable machining challenges[9,10]. During conventional mechanical material removal processes, such as traditional scratching or diamond turning, there is an increasing concern regarding the material's high susceptibility to brittle fracture[11]. This fracture-dominated removal inevitably induces severe subsurface damage, including deep micro-cracks, uncontrolled amorphous layers, and severe residual stresses[12,13]. Because high-end engineering applications dictate exceptionally strict surface integrity requirements, such machining-induced damage must be strictly minimized[14,15]. This fundamental limitation of conventional continuous cutting provides an additional drive to explore advanced manufacturing techniques capable of altering the stress state in the cutting zone. To achieve reliable ductile-regime machining in such brittle substrates, researchers need to investigate alternative kinematic strategies that can effectively suppress crack initiation and propagation[16].

VAM has been widely demonstrated to enhance the machinability of various hard-

to-cut materials[17–23]. However, monocrystalline silicon presents a uniquely complex challenge: its primary mechanism for plastic deformation relies heavily on stress-induced phase transformations[24,25]. Consequently, the exact fundamental tribological interactions and intermittent cutting mechanisms by which VAM promotes ductile material removal in silicon are still ill understood.

A substantial body of rigorous and highly valuable research has investigated the VAM of monocrystalline silicon. For instance, Li et al. investigated rotary ultrasonic micro-machining of silicon, revealing that ultrasonic vibration significantly reduces cutting forces and edge chipping by facilitating intermittent contact and material removal[26]. Similarly, Ansary et al. explored the parametric effects of different vibration modes in micro-grooving, demonstrating that horizontal vibration minimizes surface roughness while vertical vibration reduces thrust forces, though it can increase edge damage[27]. To isolate the intermittent impact effect, Huang and Yan utilized cyclic nanoindentation to simulate ultrasonic vibration cutting (UVC), discovering that cyclic loading promotes silicon amorphization and decreases the threshold for phase transitions (e.g., β -Sn to bc8 and r8 phases)[28]. While these studies are invaluable, the exact role of vibration amplitude and frequency in the ductile-to-brittle transition (BDT) and subsurface damage evolution is underrepresented in current literature, often limiting performance optimization in industrial settings. Most macroscopic experimental studies operating near the brittle regime lack a comprehensive analysis of the underlying SSD, often deducing the BDT solely from surface topography.

To probe the nanoscale mechanisms, researchers have turned to atomic force microscopy (AFM). Wang et al. employed AFM to investigate the material removal mechanisms involved in 3D ultrasonic vibration-assisted nanomilling (UVAN) of monocrystalline silicon[29]. Their findings indicate that material removal is governed predominantly by shear when the maximum undeformed chip thickness exceeds a critical value, whereas compressive extrusion dominates below this threshold. Furthermore, Tang et al. developed an ultrasonic elliptical vibration (UEV) stage for AFM tip-based nanofabrication, showing that UEV significantly increases machining

depth while reducing the subsurface amorphous layer[30]. However, because AFM employs extraordinarily small tip radii and ultra-low contact loads, its machining range is perpetually confined within the plastic regime. Consequently, this approach misses the critical macroscopic phenomena occurring at the BDT boundary, leaving the transition mechanism under scrutiny.

Concurrently, molecular dynamics (MD) simulations have been extensively utilized to capture transient atomic-level changes. Chen et al. conducted MD simulations of vibration-assisted nano-scratching on silicon, quantifying how vibration redistributes the anisotropic stress state and reduces the amorphous-layer thickness [31]. Similar MD approaches by Wang et al. [29] and Tang et al. [30] have provided profound insights into dislocation evolution and phase transformations. While a comprehensive summary of all MD advancements is beyond our current scope, it is indeed critical to note a shared limitation. Due to computational constraints, these MD studies are restricted to picosecond (ps) and nanometer (nm) scales, necessitating the use of artificially high vibration frequencies, often in the gigahertz (GHz) range[31–35]. Although these meticulous simulations enhance our atomic-scale understanding, they lack insights at the spatiotemporal scales of industrially viable machining strategies. Therefore, a critical gap remains in linking the macroscopic intermittent cutting mechanisms of VAM to the microstructural realities of the BDT.

To bridge the aforementioned gaps between macroscopic phenomena and nanoscale phase transformations, this study provides a detailed reference and rigorous quantification of the VAM process through instrumented nanoscratch testing. The primary objective is to uncover the fundamental thermomechanical mechanisms that suppress brittle fracture and allow the diamond tool to withstand severe wear during processing.

Through comparative experimental analysis, this study demonstrates that VAM successfully suppresses the formation of micro-cracks at applied loads where conventional machining exhibits obvious and catastrophic brittle fracture. The transition to a purely ductile material removal regime under ultrasonic modulation is

clearly evidenced by the generation of continuous, ribbon-like chips, which is a definitive hallmark of plastic deformation in nominally brittle materials.

The underlying crystallographic mechanism driving this enhanced plasticity is discussed in detail via cross-sectional Transmission Electron Microscopy (TEM). Our findings reveal a fundamental divergence in the stress-unloading phase transformation pathways. In conventional cutting, the high-pressure metallic Si-II phase predominantly transforms into amorphous silicon (a-Si) upon load release. In contrast, the cyclic loading and unloading inherent to VAM alter this pathway, causing the Si-II phase to transform into a composite structure of amorphous silicon and metastable crystalline Si-III. Crucially, these newly formed crystalline Si-III regions facilitate subsequent deformation through active dislocation propagation. This localized dislocation mobility effectively dissipates accumulated stress, thereby promoting sustained plastic flow and actively preventing the initiation of brittle fracture.

By elucidating this cyclic phase-transformation mechanism, this research provides the necessary quantification of how VAM extends the ductile-to-brittle transition boundary. Ultimately, these insights will find use in diverse and wide applications, ranging from advanced MEMS to high-power optoelectronics.

2. Experimental Methods

2.1. Material preparation and scratch setup

The scratch experiments were conducted on pristine, defect-free monocrystalline silicon wafers. Prior to testing, the initial surface morphology was measured using a nanoindenter (Bruker/Hysitron TI980), revealing an average surface roughness (R_a) of approximately 0.16 nm (Fig. 1(b)). The high crystalline quality of the initial silicon sample was further confirmed via High-Resolution Transmission Electron Microscopy (HRTEM), which displayed a perfect lattice structure without pre-existing defects (Fig. 1c). All scratch tests were performed along the [110] crystallographic direction.

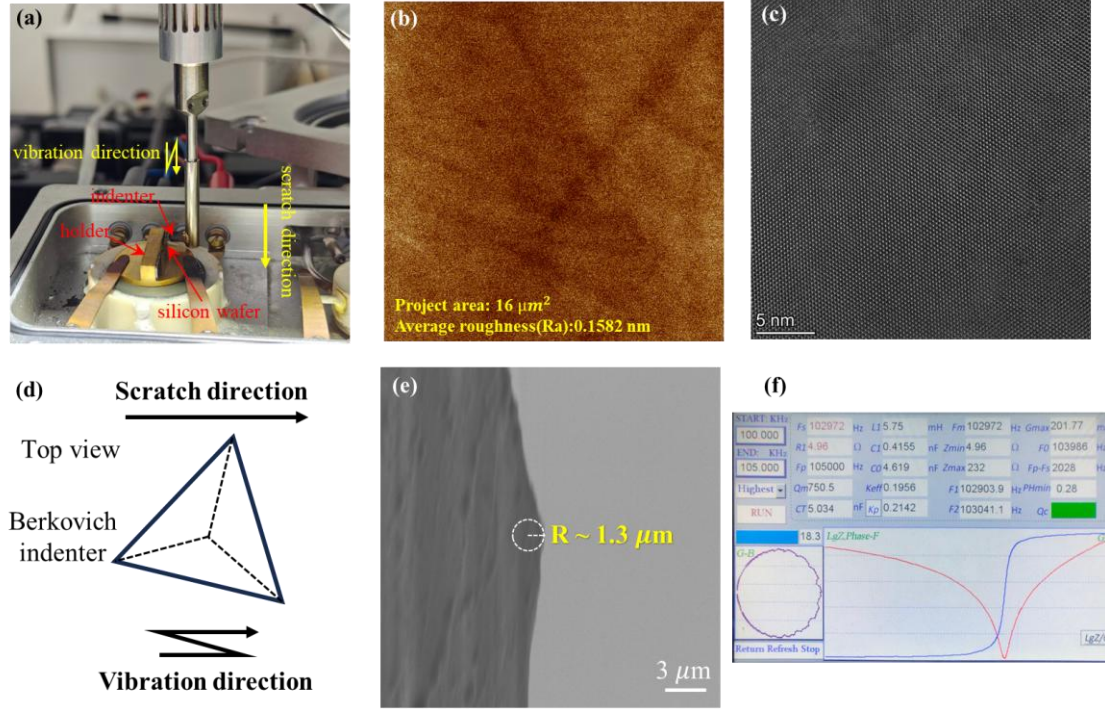


Fig. 1. Experimental setup and initial material characterization for the vibration-assisted scratch tests. (a) Configuration of the custom ultrasonic vibration device integrated with the multifunctional tribometer. (b) Initial surface morphology of the pristine monocrystalline silicon wafer. (c) HRTEM image confirming the defect-free lattice structure prior to testing. (d) Schematic of the indenter orientation relative to the [110] crystallographic scratch direction. (e) Profile of the Berkovich diamond indenter. (f) Resonant characteristics of the ultrasonic device, displaying the admittance circle and frequency sweep.

The tests were carried out using a multifunctional tribometer (MFT-5000, Rtec Instruments, USA), as illustrated in Fig. 1(a). To achieve a ramp scratch under force control, a custom-designed holder was utilized. The holder was inclined using precision shims to create a slope of approximately 1:40 relative to the scratch direction. In this configuration, the tangential force was linearly increased from 0 to 3 mN during the scratching process. A Berkovich diamond indenter with a tip radius (R) of approximately $1.3 \mu\text{m}$ was employed (Fig. 1e). Although the orientation of the indenter relative to the cutting direction is known to influence the material removal mechanism,

this study focuses specifically on the effects of vibration. Therefore, a uniform indenter orientation (shown in Fig. 1(d)) was maintained across all tests to eliminate geometric variables.

2.2. Ultrasonic vibration system and parameters

A custom-developed ultrasonic vibration device was integrated into the tribometer system to perform the vibration-assisted scratch tests. The resonant characteristics of the piezoelectric device were evaluated using a precision impedance analyzer (PV520A-S, GUOBIAO). As shown in Fig. 1(f), the admittance circle and frequency sweep results indicate a sharp resonance peak, confirming the stability of the system. The measured series resonance frequency (F_s) was 102.97 kHz, with a dynamic resistance (R_1) of 4.96 Ω and a mechanical quality factor (Q_m) of 750.5.

To accurately calibrate the vibration amplitude at the tool tip, a non-contact Laser Doppler Vibrometer (Aunion) was employed. During calibration, it was observed that due to the inherent compliance of the tool assembly, the longitudinal excitation induced a coupled vibration. Consequently, the indenter oscillated not only in the primary scratch direction but also in the normal direction (perpendicular to the contact surface). The quantified amplitudes for the conventional non-ultrasonic vibration (NUV) baseline and the three ultrasonic vibration (UV) conditions (UV-10 nm, UV-50 nm, and UV-110 nm) are summarized in Table 1.

Table 1. Kinematic parameters for the ultrasonic vibration-assisted (UV) scratching experiments.

	Scratch direction amplitude(nm)	Normal direction amplitude(nm)	f
UV-10 nm	10	0	102.9kHz
UV-50 nm	50	10	102.9kHz
UV-110 nm	110	20	102.9kHz

2.3. Characterization techniques

Following the scratch tests, comprehensive characterization was performed to evaluate both surface and subsurface integrity. The surface morphology, groove dimensions, and chip formation characteristics were examined using a high-resolution field-emission scanning electron microscope (FE-SEM, Zeiss Merlin).

To investigate the SSD and phase transformations induced by the cutting process, cross-sectional analysis was focused on the scratch region corresponding to a normal load of approximately 50 mN. Cross-sectional TEM specimens were extracted and prepared using a focused ion beam (FIB) system (Helios 600i, FEI). To minimize ion-beam-induced amorphization and ensure uniform specimen thickness, a rigorous multi-stage milling protocol was implemented. Initial thinning was performed at an accelerating voltage of 30 kV, with the beam current progressively reduced from 0.79 nA to 80 pA. During this stage, a $\pm 1^\circ$ specimen tilt was applied to mitigate curtaining effects until a target thickness of approximately 100 nm was achieved. This was followed by a final low-energy polishing step at 5 kV (15 pA) with a $\pm 3^\circ$ tilt for 3 minutes per angle to remove any residual surface damage layer introduced by the high-energy gallium ions.

The prepared cross-sectional specimens were subsequently examined using a transmission electron microscope (Talos F200X, FEI) operating at 200 kV. This allowed for high-resolution imaging (HRTEM) and selected area electron diffraction (SAED) analysis to accurately characterize the subsurface microstructures and identify specific polymorphic phase transitions within the monocrystalline silicon.

3. Results and Discussion

3.1. Kinematic analysis and force responses under force-controlled VAM

To elucidate the contact mechanics during the vibration-assisted scratching process, the tool kinematics must be quantitatively defined. The instantaneous displacement $x(t)$ and velocity $x'(t)$ of the tool in the cutting direction are expressed as:

$$x(t) = Vt + A \sin(2\pi ft) \quad (1)$$

$$x'(t) = V + 2\pi fA \cos(2\pi ft) \quad (2)$$

where V is the nominal cutting velocity, A is the vibration amplitude, and f is the vibration frequency. The fundamental mechanism of VAM relies on the intermittent separation between the tool and the workpiece. This separation is only physically possible if the nominal cutting velocity is lower than the maximum vibration velocity of the tool. This threshold is defined as the critical velocity V_{crit} :

$$V_{crit} = 2\pi fA \quad (3)$$

To quantify the degree of this intermittent cutting, two key dimensionless parameters are utilized: the Horizontal Speed Ratio (HSR) and the Duty Cycle (DC_1). The HSR represents the ratio of the nominal cutting speed to the maximum vibration speed:

$$HSR = \frac{V}{V_{crit}} = \frac{V}{2\pi fA} \quad (4)$$

A lower HSR indicates a higher relative vibration velocity, which typically facilitates better chip evacuation and cooling. The Duty Cycle, DC_1 , defines the fraction of the total vibration period ($T = 1/f$) during which the tool is in physical contact with the workpiece material. It is calculated based on the tool entry time (t_1) and separation time (t_2):

$$DC_1 = \frac{t_2 - t_1}{T} = f(t_2 - t_1) \quad (5)$$

The separation time t_2 is determined by solving for the point where the instantaneous velocity of the tool equals the nominal cutting speed:

$$t_2 = \frac{\arccos(-V/2\pi fA)}{2\pi f} \quad (6)$$

Based on these kinematics, V_{crit} , HSR , and DC_1 were calculated for the three amplitude conditions (10 nm, 50 nm, and 110 nm) and are summarized in Table 2.

Table 2. Variations in critical velocity, speed ratio, and duty cycle across the tested UV scratching amplitudes.

	$V_{crit}(\text{mm/s})$	HSR	$t_2(\mu\text{s})$	DC_1
UV-10 nm	6.465	0.00144	2.4294	0.2502
UV-50 nm	38.795	0.00029	2.4276	0.2500
UV-110 nm	85.343	0.00013	2.4274	0.2499

Unlike conventional displacement-controlled scratch tests, the present study utilized a force-controlled methodology where the lateral force (in the scratch direction) was linearly increased from 0 to 3 mN across all test groups (Fig. 2(a)). Consequently, the strain rate could not be explicitly controlled; however, post-experimental calculations determined the average indenter velocity along the scratch direction to be $V = 0.0093\text{mm/s}$.

Because the lateral resistance was identical (0 to 3 mN) for all conditions, the primary variable of interest was the system's response in the normal direction. Experimental data indicates that only the UV-110 nm condition exhibited a significant increase in the accommodated normal load. In other words, for a given lateral cutting resistance, the 110 nm amplitude condition allowed the tool to sustain a larger normal force, which fundamentally translates to a significantly reduced coefficient of friction (COF).

According to the kinematic calculations in Table 2, the duty cycles for all three UV groups are nearly identical (approximately 0.25). Therefore, the standard 1D VAM theory, which attributes friction reduction primarily to a decreased duty cycle cannot explain the distinct COF reduction observed in the UV-110 nm group. Instead, this phenomenon is more likely attributable to the oscillation of the tool in the normal direction. At an amplitude of 110 nm, the normal-direction vibration component becomes sufficient to induce actual physical separation from the contact surface for extended durations within each cycle, thereby breaking the continuous tribological contact and lowering the macroscopic friction coefficient.

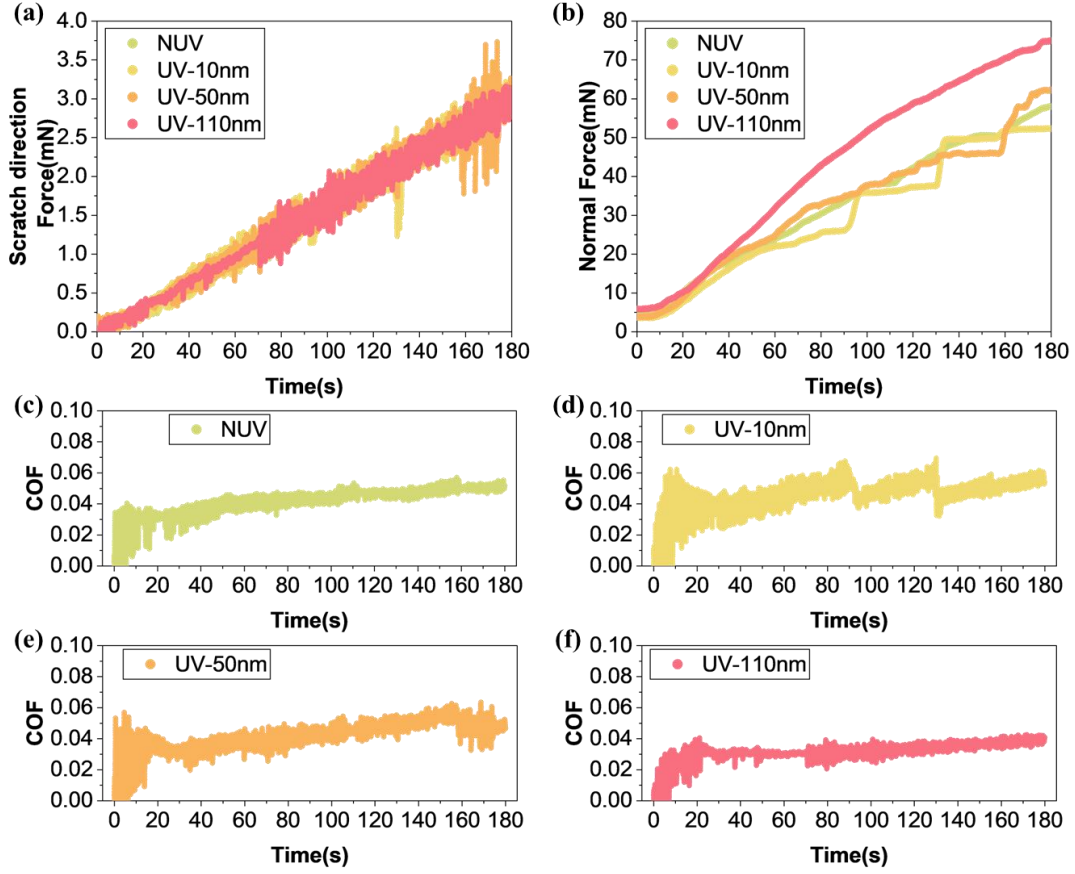


Fig. 2. Force responses and COF during force-controlled scratching of monocrystalline silicon. (a) Linearly applied scratch force. (b) Accommodated normal force. Quantification of COF for (c) NUV, (d) UV-10nm, (e) UV-50nm, and (f) UV-110nm.

Conversely, for the 10 nm and 50 nm conditions, the amplitude is too small to overcome the elastic recovery of the silicon and the compliance of the system in the normal direction. Because the applied frequency is exceptionally high and the amplitudes are minimal, the variations in amplitude have a negligible impact on both HSR and DC_1 . Consequently, the results demonstrate that under high-frequency, small-amplitude loading conditions, the efficacy of 1D VAM in reducing the friction coefficient is inherently limited unless the amplitude threshold required for normal-direction surface separation is exceeded.

3.2. Surface morphology and ductile-to-brittle transition

To elucidate the material removal mechanisms and the ductile-to-brittle transition of monocrystalline silicon, the surface morphologies of scratches generated under conventional (NUV) and UV conditions were systematically compared. Figure 3-6

presents the SEM micrographs of the scratch grooves along the cutting direction, categorized into four distinct regions (I to IV) corresponding to increasing normal loads.

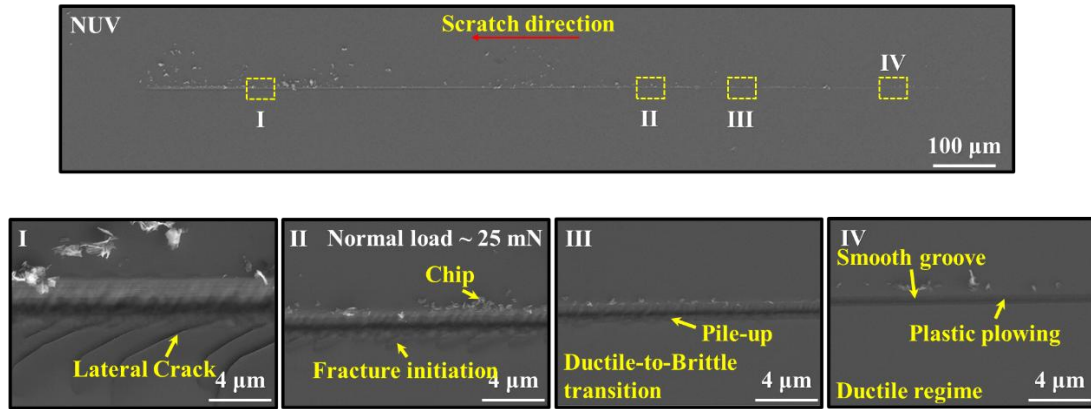


Fig. 3. SEM morphology of the scratch groove on monocrystalline silicon under conventional cutting (NUV). Insets I, II, III, and IV show the magnified views of the corresponding local regions marked in the full scratch track.

In the conventional NUV process (Fig. 3), a typical DBT sequence is clearly observed. In the initial stage with low normal load (Region IV), material is removed purely through plastic deformation, leaving a smooth groove indicative of the ductile regime. However, as the scratch progresses and the normal load reaches approximately 25 mN (Region II), fracture initiation occurs, characterized by the sudden appearance of localized chipping and pile-up (Region III). Beyond this critical threshold, as the load continues to increase towards the end of the scratch (reaching up to approximately 50 mN), the material removal mode completely transitions to the brittle regime (Region I). This terminal region is dominated by the continuous generation and extensive propagation of severe lateral cracks, indicating that under static loading, the accumulated stress rapidly and irreversibly exceeds the fracture toughness of silicon.

In stark contrast, the introduction of ultrasonic vibration fundamentally alters the stress distribution and crack propagation dynamics. The most striking observation is that the UV groups do not merely delay fracture at the 25 mN threshold; rather, they demonstrate a remarkable capability to suppress lateral crack formation entirely throughout the remainder of the scratch. Even as the normal load peaks at 50 to 70 mN (varying by amplitude group) at the end of the scratch, the catastrophic lateral cracking ubiquitous in the NUV high-load regime is conspicuously absent across all UV conditions.

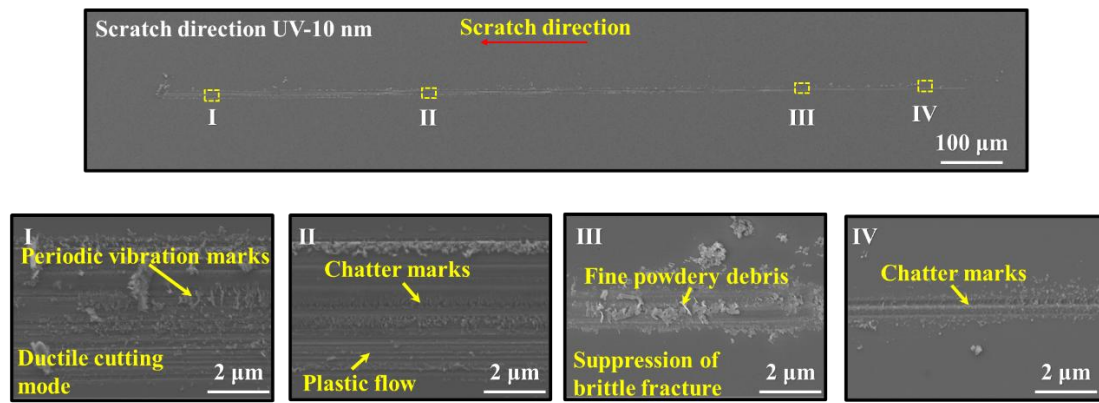


Fig. 4. SEM morphology of the scratch groove on monocrystalline silicon under vibration-assisted cutting with an amplitude of 10 nm (UV-10nm). Insets I, II, III, and IV show the magnified views of the corresponding local regions marked in the full scratch track.

For the UV-10 nm condition (Fig. 4), despite reaching high terminal loads, the severe lateral cracks seen in NUV are effectively eliminated. The surface exhibits a ductile cutting mode (Region I) with periodic vibration marks and chatter marks (Regions II and IV). The debris primarily consists of fine powder (Region III), suggesting that the material is removed through highly localized, high-frequency micro-fractures and plastic flow, successfully preventing large-scale cleavage even at extreme loads.

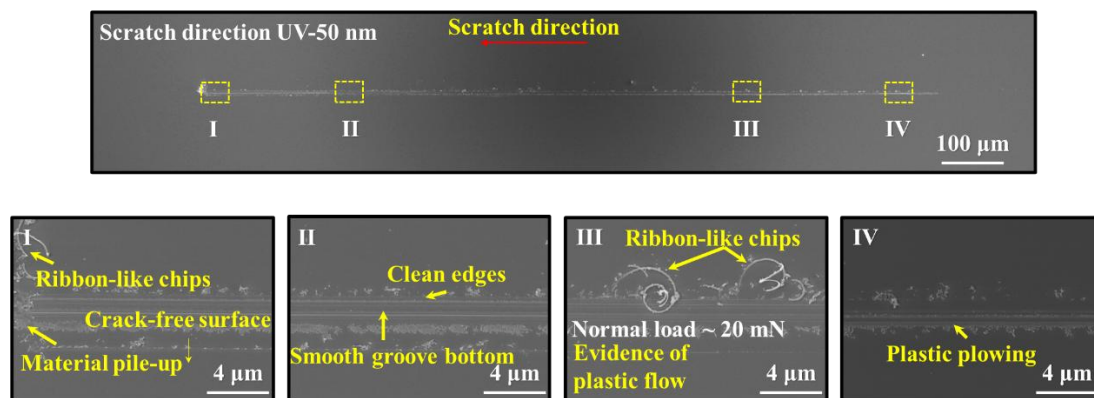


Fig. 5. SEM morphology of the scratch groove on monocrystalline silicon under vibration-assisted cutting with an amplitude of 50 nm (UV-50nm). Insets I, II, III, and IV show the magnified views of the corresponding local regions marked in the full scratch track.

The optimal surface integrity is achieved at a vibration amplitude of 50 nm (Fig.

5). The SEM images reveal a highly stable plastic deformation mechanism that persists through the entire high-load regime. Even at the maximum normal loads near the end of the scratch (Regions I and III), the groove maintains a completely crack-free surface with clean edges and a remarkably smooth bottom (Region II). Most notably, the material is removed in the form of continuous, ribbon-like chips. The sustained formation of these continuous chips under loads of 50~70 mN is a direct physical manifestation of enhanced material plasticity. The intermittent tool-workpiece contact at this specific amplitude allows for sufficient stress relaxation, thereby preventing the critical stress concentration required for crack initiation regardless of the high macroscopic normal force.

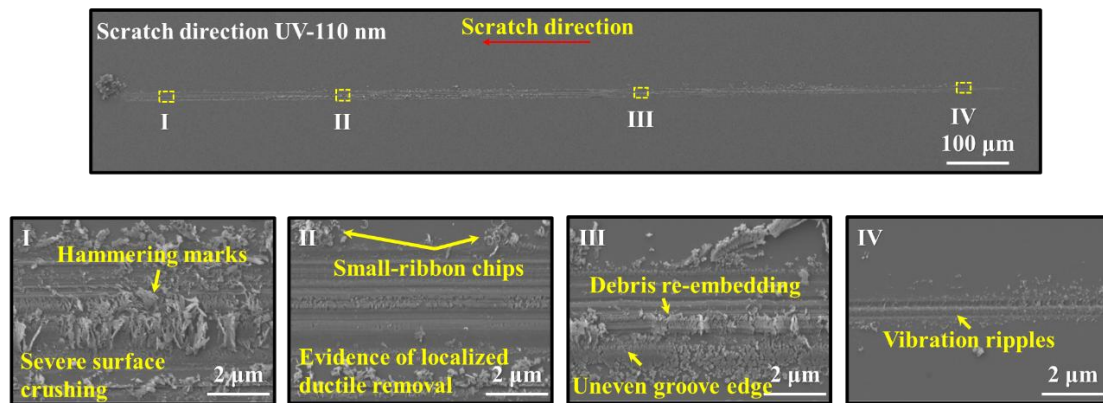


Fig. 6. SEM morphology of the scratch groove on monocrystalline silicon under vibration-assisted cutting with an amplitude of 110 nm (UV-110nm). Insets I, II, III, and IV show the magnified views of the corresponding local regions marked in the full scratch track.

However, when the vibration amplitude is further increased to 110 nm (Fig. 6), while massive lateral cracks are still suppressed, the kinematic advantages of VAC are compromised by excessive impact energy. The surface morphology deteriorates significantly at high loads, evidenced by severe surface crushing and hammering marks (Region I). The high-amplitude tool trajectory causes uneven groove edges and debris re-embedding (Region III). Although small-ribbon chips and localized ductile removal are still visible (Region II), the presence of severe crushing indicates that the cutting mechanism has shifted from smooth intermittent shearing to a detrimental hammering effect, which degrades the overall surface quality despite the absence of traditional brittle cleavage.

3.3. Subsurface damage and phase transformation mechanisms

To elucidate the subsurface deformation and phase transformation mechanisms

governing the cutting process, cross-sectional TEM samples were extracted from the NUV and UV(10 nm) groups at a normal load of ~50 mN. Furthermore, to investigate the physical origin of the ribbon-like curled chips observed under higher amplitudes, an additional TEM sample was prepared from the UV(50 nm) group at the exact location where these continuous chips first emerged (normal load ~20 mN).

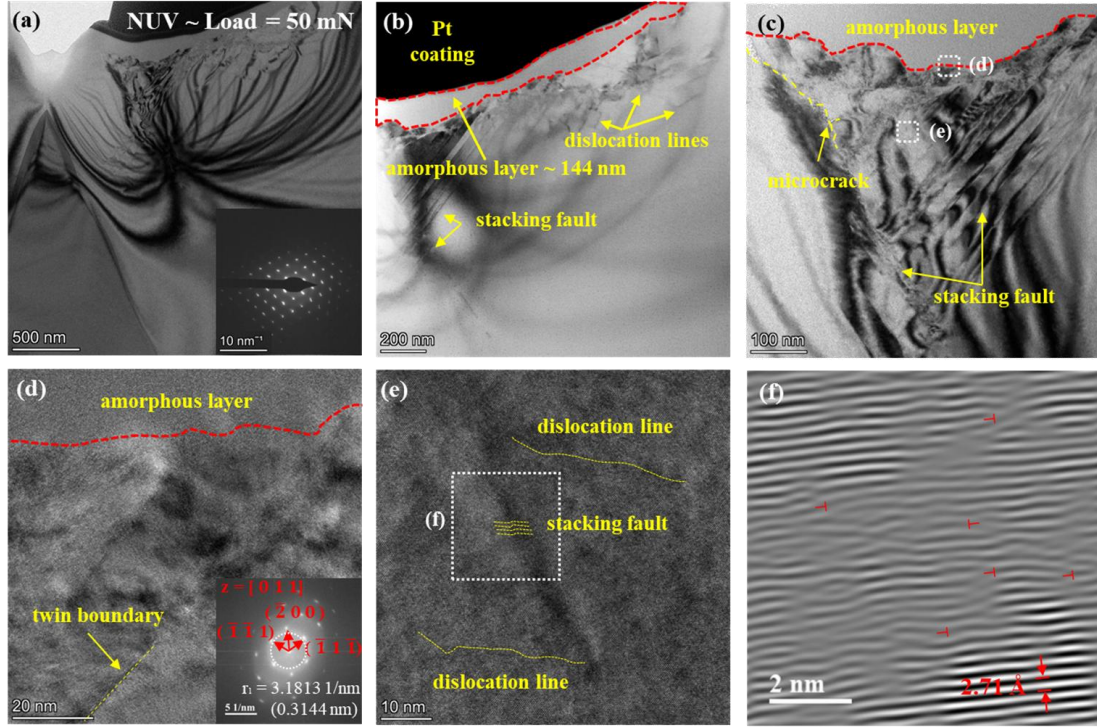


Figure 7. Cross-sectional TEM analysis of the NUV scratch at a normal load of 50 mN. (a) Bright-field image of the subsurface damage. (b) Magnified view of the amorphous layer and defects. (c) Localized view of microcracks and stacking faults. (d) HRTEM image of the amorphous layer and twin boundary. (e) HRTEM image of dislocation lines. (f) IFFT image detailing the dislocations.

Figure 7 presents the cross-sectional TEM analysis of the NUV scratch at 50 mN. A severe structural degradation is evident, characterized by a prominent amorphous layer with a thickness of approximately 144 nm (Fig. 7(b)). In monocrystalline silicon, deformation within amorphous regions is highly localized. This lack of structural order prevents the distribution of strain, leading to rapid and severe localized stress concentration. Once this accumulated stress exceeds the local fracture toughness, the material fails via brittle fracture. This mechanism is directly supported by the observation of microcracks propagating beneath the amorphous layer (Fig. 7(c)), alongside localized stacking faults and twin boundaries.

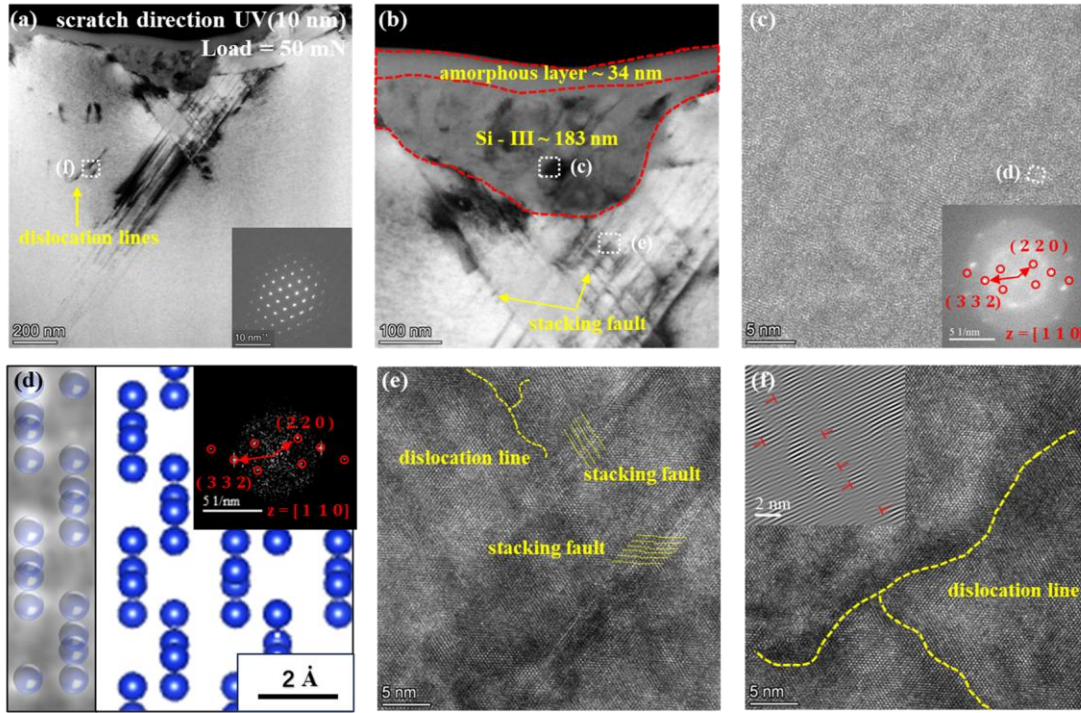


Figure 8. Cross-sectional TEM analysis of the UV (amplitude = 10 nm) scratch at a normal load of 50 mN. (a) Bright-field image along the scratch direction. (b) Magnified view of the amorphous layer and Si-III phase. (c) HRTEM image of the Si-III region. (d) Atomic model and corresponding FFT pattern. (e) HRTEM image showing dislocation lines and stacking faults. (f) IFFT image of the dislocations.

In contrast, the application of vibration assistance fundamentally alters the subsurface microstructure. Under UV (10 nm) conditions at the same 50 mN load (Figure 8), the thickness of the amorphous layer is drastically reduced to ~34 nm. More significantly, a thick layer (~183 nm) of the metastable Si-III crystalline phase emerges directly beneath the amorphous zone (Fig. 8(b)). The presence of this phase is confirmed by the HRTEM and FFT (Fig. 8(d)).

HRTEM and FFT analyses were conducted to accurately index the crystallographic structure within this transformation zone. The observations reveal that the region is not monocrystalline; rather, it comprises nanocrystalline domains with varying orientations, a structural evolution driven by the high-stress environment during the cutting process. Specifically, the generated FFT patterns exhibit distinct diffraction spots with measured interplanar spacings of 0.232 nm and 0.140 nm. These measurements correspond closely to the (220) and (332) crystallographic planes of the metastable Si-III (BC8) phase, which have theoretical spacings of 0.235 nm and 0.141 nm, respectively[36]. Furthermore, the measured interplanar

angle between these planes is 27.4° . Although this value deviates slightly from the theoretical angle of 25.2° , this geometric discrepancy is physically consistent with the severe lattice distortion and high residual stresses induced by the mechanical interaction. The preservation of such a substantial volume of the metastable Si-III phase post-unloading serves as a distinct microstructural characteristic of this specific vibration-assisted cutting mode.

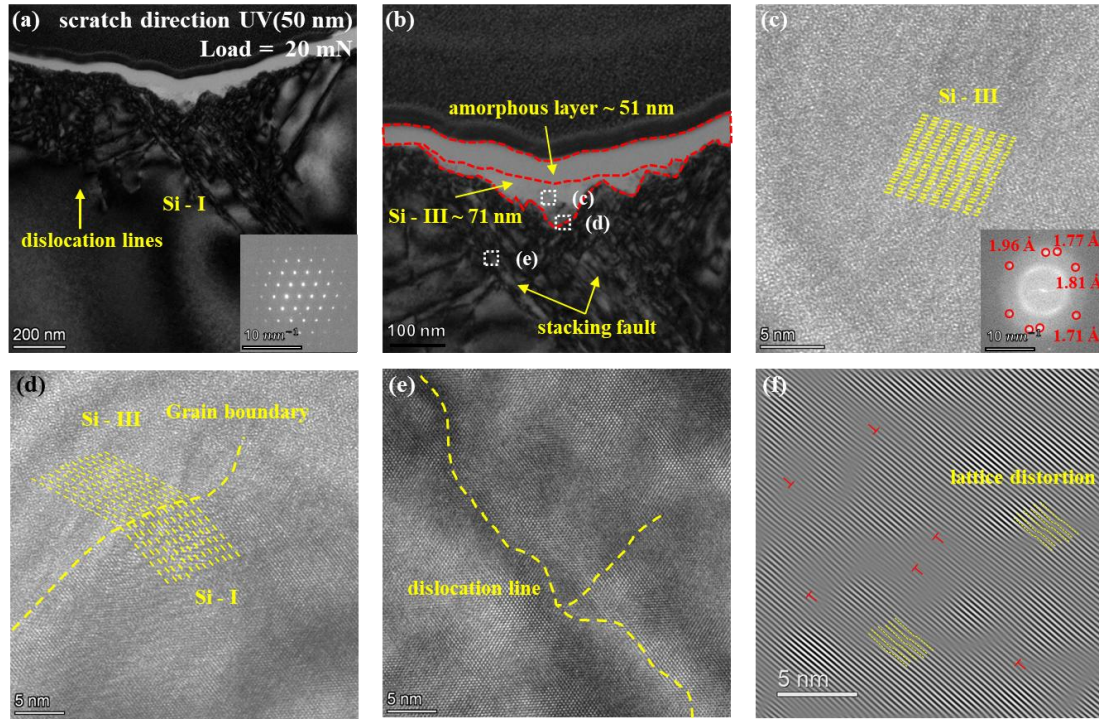


Figure 9. Cross-sectional TEM analysis of the UV (amplitude = 50 nm) scratch at a normal load of 20 mN. (a) Bright-field image of the subsurface damage. (b) Magnified view of the amorphous and Si-III layers. (c) HRTEM image and FFT pattern of the Si-III phase. (d) HRTEM image of the grain boundary between Si-III and Si-I. (e) HRTEM image of a dislocation line. (f) IFFT image displaying the lattice distortion.

The mechanism enabling ductile material removal is further clarified by the UV (50 nm) sample taken at ~ 20 mN (Figure 9), corresponding to the onset of ribbon-like chip formation. In this region, an amorphous layer of ~ 51 nm and a distinct Si-III layer of ~ 71 nm are observed (Fig. 9(b)), along with clear grain boundaries separating the Si-I and Si-III phases (Fig. 9(d)). The critical difference between the NUV and UV conditions lies in the mobility of crystallographic defects. While amorphous deformation causes stress concentration and fracture, crystalline phases, including the unstable Si-III phase, allow for the extensive generation and mobility of dislocations

and stacking faults (Fig. 9(e, f)).

During the scratching process, dislocation lines typically propagate downwards and in the direction opposite to the tool movement[37]. The high-frequency intermittent contact inherent to the UV process creates periodic stress loading and unloading cycles. The data indicates that the stress-unloading regions facilitate the phase transformation from the highly stressed amorphous state or Si-I into the Si-III crystalline phase. Because defect mobility is significantly enhanced in this crystalline state, the deformation can propagate continuously via dislocation slip rather than accumulating as localized stress. This continuous dissipation of strain energy through dislocation movement in the Si-III phase suppresses microcrack initiation and accommodates extensive plastic strain, thereby explaining the macroscopic observation of obvious plastic material removal and the formation of continuous, ribbon-like chips in the UV groups.

3.4. Mechanism of ultrasonic vibration-assisted scratching

Based on the microstructural observations and kinematic analysis, a comprehensive schematic model of the material removal and phase transformation mechanisms during conventional and ultrasonic vibration-assisted scratching is proposed in Figure 10. In conventional continuous scratching (Figure 10, bottom right), the indenter maintains constant engagement with the monocrystalline silicon (Si-I) at a uniform velocity v . This continuous interaction generates a sustained hydrostatic pressure field beneath the tool, driving the bulk phase transformation from the diamond-cubic Si-I to the metallic β -tin (Si-II) phase. As the tool advances, the continuous release of this pressure typically results in the formation of a thick, uniform layer of amorphous silicon (α -Si) on the machined surface.

In contrast, the ultrasonic VAM process introduces a highly dynamic, intermittent cutting cycle characterized by periodic tool-workpiece separation. As illustrated in steps 1 through 3 of Figure 10, a single high-frequency vibration cycle consists of distinct kinematic phases governed by the relative tool velocity $x'(t)$. During the initial engagement phase (t_1), the indenter penetrates the material, initiating localized hydrostatic pressure and the formation of a primary Si-II zone. At maximum penetration (t_2 , where $x'(t) = 0$), the stress state peaks, defining the maximum extent of the phase transformation zone for that specific cycle. Subsequently, during the retraction phase (t_3), the tool physically separates from the workpiece. This critical separation induces a rapid, localized pressure release, prompting the unstable Si-II phase to transform into α -Si and other metastable structures.

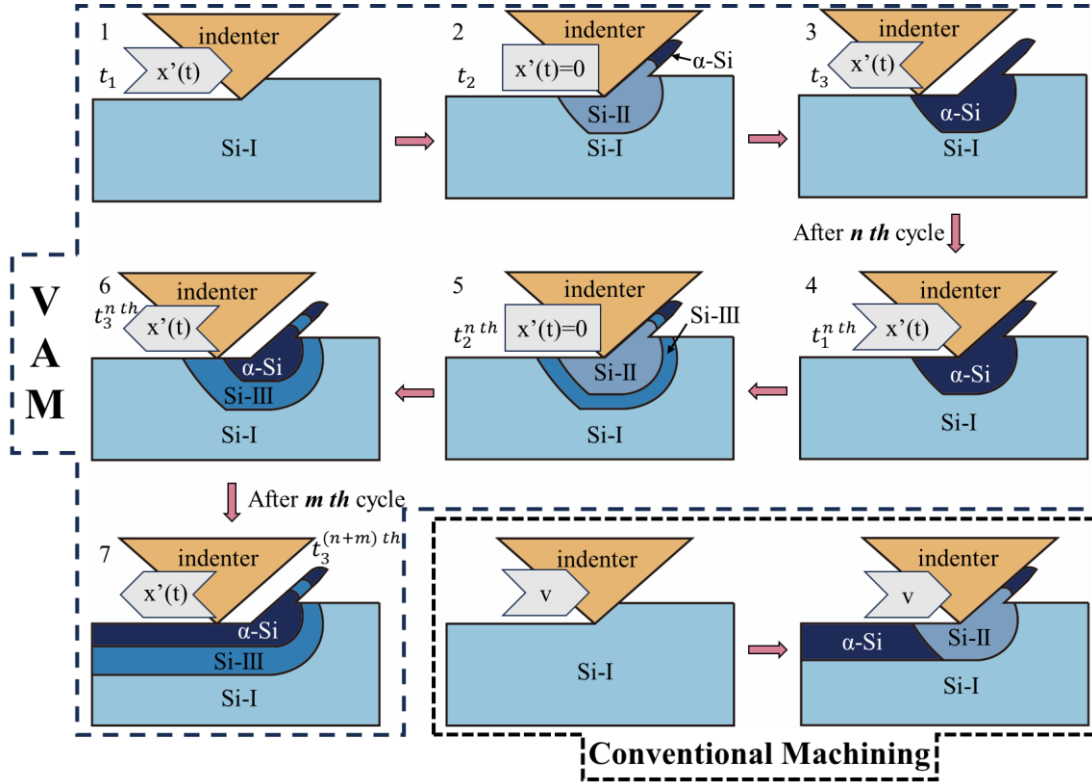


Fig. 10. Comprehensive schematic model illustrating the material removal and phase transformation mechanisms in monocrystalline silicon. The main sequence (larger dashed box) demonstrates the vibration-assisted scratching, the bottom-right section (bottom right dashed box) illustrates conventional continuous scratching at a constant velocity.

As the cutting process progresses through multiple overlapping cycles (e.g., the n -th and m -th cycles shown in steps 4 to 7), the material experiences a continuous cyclic loading and unloading effect. Previous static models, which rely on steady-state stress field assumptions, generally miss this high-frequency stress relaxation behavior. The dynamic stress relief inherent in VAM forces the silicon to undergo rapid, sequential pressure variations. This specific thermodynamic environment facilitates the formation and preservation of the metastable Si-III phase, which is observed sandwiched between the top α -Si layer and the Si-I bulk. This mechanism is consistent with previous cyclic indentation studies [28,38,39], which have demonstrated that repeated loading and unloading alters the phase transformation pathways of monocrystalline silicon, preferentially driving the structural transition towards the Si-III phase upon pressure release rather than solely forming amorphous silicon. Furthermore, the periodic separation between the tool and the workpiece fundamentally alters the tribological interface by breaking the continuous friction vector. This intermittent contact reduces time-averaged frictional forces and prevents localized

tensile stresses from accumulating to the critical threshold required for brittle cleavage. Consequently, the material is removed primarily through the plastic flow of the high-pressure Si-II phase, which facilitates ductile-regime machining and preserves the structural integrity of the subsurface.

4. Conclusions

This study investigated the material removal mechanisms and subsurface damage evolution of monocrystalline silicon during in-plane ultrasonic UV scratching. By correlating macroscopic kinematic force responses with nanoscale microstructural characterizations, the fundamental mechanisms governing the ductile-to-brittle transition and phase transformations under high-frequency intermittent cutting were elucidated. The main conclusions are drawn as follows:

1. Extension of the ductile regime and surface enhancement: Ultrasonic vibration significantly suppresses the initiation and propagation of lateral microcracks, leading to a drastic improvement in the machined surface quality. Compared to conventional NUV scratching, which exhibits catastrophic brittle fracture and severe surface degradation at normal loads above 25 mN, UV scratching at an optimal amplitude of 50 nm maintains a purely ductile removal mode up to 70 mN. This enhanced plasticity results in a pristine, crack-free surface topography, macroscopically evidenced by the formation of continuous, ribbon-like chips
2. Kinematic effects on interfacial friction: The reduction of macroscopic friction during UV scratching is highly sensitive to the applied kinematic parameters. Under small amplitudes and a duty cycle of approximately 0.25, the process exhibits a negligible effect on friction reduction, which is hypothesized to be a result of insufficient excitation energy at the tool-workpiece interface. While increasing the nominal amplitude to 110 nm yields a significant decrease in both the friction force and the COF, this apparent reduction may be primarily attributed to an unintended 20 nm vertical vibration component that induces physical separation between the tool and the substrate. Consequently, the isolated effects of 1D kinematics on tribological behavior in this regime remain complex, and further investigation is required to fully decouple these interfacial mechanisms.
3. Modulation of phase transformation pathways: Cross-sectional TEM analysis reveals that the high-frequency intermittent loading and unloading cycles fundamentally alter the thermodynamic pathways of silicon. Unlike NUV

cutting, which generates a thick, highly stressed amorphous layer (α -Si) prone to cracking, UV scratching facilitates dynamic stress relief. This rapid pressure release drives the high-pressure Si-II phase to transform into a distinct layer of metastable crystalline Si-III, significantly reducing the thickness of the residual amorphous layer.

4. Dislocation-mediated plasticity: The preservation of the metastable Si-III phase is the critical microstructural driver for the enhanced machinability of silicon. Unlike the amorphous phase, the crystalline Si-III structure accommodates extensive plastic strain through active dislocation propagation and stacking fault mobility. This continuous dissipation of strain energy prevents localized stress concentrations, thereby inhibiting brittle cleavage and enabling stable ductile material removal.

In summary, this work provides a comprehensive mechanistic understanding of how high-frequency intermittent contact modifies the thermomechanical state and crystallographic evolution of monocrystalline silicon. These findings offer critical theoretical guidance for optimizing vibration-assisted machining parameters to achieve high-efficiency, low-damage manufacturing of hard and brittle semiconductor materials.

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